

Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

July 10, 2026

1 Roadmap and rigour status

This lab-book tracks the report-facing slice of the registry in `argument/`. The north star is `op-classical`; its status is *open*, and the status-ledger anchor is 1. The definitions used by the registry are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`. This report references those identifiers; it does not introduce parallel definitions.

The rigorous line is deliberately narrow. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard, the only registry results reproduced as rigorous are the seven `af`-validated entries 1, 2, 3, 4, 5, 6, and 7. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is also status-sensitive. The inherited campaign distinguishes the realizable-family linear law from the worst-case `sqrt`-envelope: the linear relation is the mechanism seen in the reduction record, while the `sqrt` scale is the stability target. The sharp-exponent obstruction is `ex-hume`, anchored at 1, but it is still `proved-mod-audit` here rather than one of the rigorous results. The bridge between the signed and stochastic pictures is the validated result 1; crossing between those pictures should cite that registry id rather than re-normalising notation locally.

2 The signed-stochastic bridge

Lemma 1 (Registry contract for lem-classical-equiv). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with $\|Q^2 - Q\| \leq \eta$ gives $P = \text{theta}(2Q - 1)$ signed affine retraction with $\|P - Q\| \leq C \eta$ and neg mass $\delta \leq C \eta$, and conversely row-normalising p_{i^+} gives Q with $\|P - Q\| \leq 2 \delta$, $\|Q^2 - Q\| \leq 6 \delta + 4 \delta^2$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.

Role in the chain. This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

3 Height collapse

Proposition 2 (Registry contract for obs-height-collapse). *Height collapse: for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v (height $H = \text{dist}_1(p_v, \text{conv } W)$ maximal among rows), the invisible mass σ_v and the row negative mass ν_v satisfy $H * (1 - \sigma_v) \leq \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 19-node adversarial tree with root **validated** and taint 19/19 clean. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.

Role in the chain. This result explains why invisible mass alone does not finish the Kernel/(EX) route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

4 Mass split bookkeeping

Lemma 3 (Registry contract for `lem-mass-split`). *Mass split: for an exact signed idempotent P and any row index v , writing $a_j = P_{\{vj\}}$, $a_j^{+} = \max(a_j, 0)$, $a_j^{-} = \max(-a_j, 0)$, and $nu_v = \sum_j a_j^{-}$, one has $\sum_j a_j^{+} = 1 + nu_v$.*

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a clean 9-node adversarial tree with root `validated` and taint 9/9 `clean`. The export is `proofs/lem-mass-split/export.md`.

Role in the chain. This is the mass bookkeeping dependency factored out of the first `conj-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

5 Convex outsourcing

Lemma 4 (Registry contract for lem-residual-lower). *Convex outsourcing: let C be the convex hull of finitely many points of R^n , and suppose $p = \sum_{j=1}^m c_j p_j + (1 - s) q$ with points p_j, q in R^n , coefficients $c_j \geq 0$, $s = \sum_{j=1}^m c_j < 1$, and $\text{dist}_1(p_j, C) \leq \text{dist}_1(p, C)$ for every j ; then $\text{dist}_1(p, C) \leq \text{dist}_1(q, C)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root validated, and taint 36/36 clean. The export is proofs/lem-residual-lower/export.md.

Role in the chain. This is a frame-free convex geometry dependency for conj-halo-collapse. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

6 Residual distance bound

Lemma 5 (Registry contract for lem-residual-upper). *Residual distance bound: let C be the convex hull of finitely many points of $\mathbb{R}^n$, let $b_1..b_M, c_1..c_N \geq 0$ with $m = \sum_j b_j - \sum_k c_k > 0$, let p_j, r_k be points of $\mathbb{R}^n$ with $q = (\sum_j b_j p_j - \sum_k c_k r_k) / m$, and let $D_k \geq 0$ satisfy $\|x - r_k\|_1 \leq D_k$ for all x in C and each k ; then $m * \text{dist}_1(q, C) \leq \sum_j b_j * \text{dist}_1(p_j, C) + \sum_k c_k * D_k$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root validated, and taint 52/52 clean. The export is proofs/lem-residual-upper/export.md. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.

Role in the chain. This is the companion frame-free convex geometry dependency for conj-halo-collapse. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, now itself validated in 6.

7 Halo-robust height collapse

Lemma 6 (Registry contract for conj-halo-collapse). *Halo-robust height collapse:*
*for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v of height H , let σ be the invisible mass of v , σ_g the halo-robust invisible mass (the positive coefficient mass v places on rows at $\ell-1$ distance $> \tau/4$ from $\text{conv } W$, $\tau = \sqrt{\delta}$), and ν_v the row negative mass; then $H * (1 - \sigma_g) \leq (\sigma - \sigma_g) * \tau/4 + \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 20-node adversarial tree, root `validated`, taint 20/20 clean (run 2 on the factored workspace). The export is `proofs/conj-halo-collapse/export.md`. The three af-validated dependencies 3, 4, and 5 are imported as externals, so the root rests only on validated or cited inputs.

Elevation history. Run 1 (pre-factoring) ballooned past the node cap (49 live nodes over the cap of 40) and was aborted as a structure signal, not a mathematical refutation: the mass-split book-keeping and the residual-distance estimates were being re-derived inline across siblings. Factoring those into the three registry sub-lemmas above cured it; run 2 closed at 20 nodes.

Role in the chain. This bridge refines the validated collapse bound 2 by pricing recipients inside the $\tau/4$ -halo of the visible hull at their actual distance rather than the worst case H , so the bound stays non-vacuous even when the raw invisible mass exceeds one through self or halo mass (1). Together with a halo-robust cap $\tilde{\sigma}_g \leq 1 - c$ for hidden top vertices it would give $H = O(\tau)$, the height cap of the Kernel Conjecture 1. That cap is OPEN: nothing in this section promotes the kernel chain.

8 Factorization bound

Lemma 7 (Registry contract for lem-factorization). *Factorization bound: let P be an exact signed idempotent (square real matrix with $P^2 = P$ and all row sums equal to 1), let $\delta(P) = \max_i \sum_j \max(-P_{ij}, 0)$, let $k = \text{rank}(P)$, and let $U = (u_1, \dots, u_k)$ be an actual-row basis of P (the rows $p_{\{u_1\}}, \dots, p_{\{u_k\}}$ of P form a basis of the row space of P) whose Gram volume satisfies $\text{Vol}(U) \geq (1/2) * \text{Vol}_{\max}(P)$, where $\text{Vol}_{\max}(P)$ is the maximum Gram volume over all actual-row bases of P ; define coordinates $a_t(j)$ by $p_j = \sum_t a_t(j) p_{\{u_t\}}$, and for each pivot index s in $\{1, \dots, k\}$ set $\beta_s(j) = P_{\{u_s\}j}$, $\lambda_s(j) = 1 - a_s(j)$, $\mu_s(j) = \sum_{t \neq s} \max(-a_t(j), 0)$, $\sigma_s(j) = \sum_{t \neq s} \max(a_t(j), 0)$, $E_s(j) = \max(\mu_s(j) - \lambda_s(j), 0)$, $\Phi_s(U) = \sum_j \max(\beta_s(j), 0) * E_s(j)$, and $S*_s(U) = \sum_j \max(\beta_s(j), 0) * (\sigma_s(j) + 2 * \max(-\lambda_s(j), 0))$; then for every pivot s , $S*_s(U) \leq 2 * \Phi_s(U) + 6 * \delta(P)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records an 11-live-node adversarial tree (12 total nodes, 1 archived), root validated, taint 12/12 clean, closed in run 1 after five rounds. The export is proofs/lem-factorization/e. The contract was narrowed to the single inequality before seeding; the inherited tightness claim for the constants (2,6) was deliberately not elevated and remains proved-mod-audit material in the registry shard body.

Role in the chain. This is the composition link of the (EX) program: any universal chart bound $\max_s \Phi_s(U_0) \leq C_0 \delta(P)$ over the $\theta = 1/2$ class now composes rigorously to the class-score constant $C_{sf} = 2C_0 + 6$. The (EX) input itself (1) remains a conjecture: certified numerical evidence supports a plateau near $C_0 = 2$, but nothing in this section promotes it.

9 Zero-sum triangle bound

Lemma 8 (Registry contract for `lem-zerosum-triangle`). *Zero-sum triangle bound:*
let w and v be vectors in $\mathbb{R}^d$ with v having coordinate sum zero ($\sum_l v(l) = 0$);
write $n(x) = \sum_l \max(-x(l), 0)$; then $n(w - v) \leq n(w) + n(v)$.

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 10-node adversarial tree, root `validated`, taint 10/10 `clean`, closed in run 1 after three rounds. The export is `proofs/lem-zerosum-triangle/export.md`. The run-1 MISSING-fact challenge isolated the load-bearing hypothesis: the identity $\sum_l \max(v(l), 0) = n(v)$ holds exactly because v has zero coordinate sum; for general v only the pointwise inequality survives.

Role in the chain. First factored dependency of 10: it was split out of the `lem-fan-payment` workspace after two balloon aborts forced a linear proof chain into cross-sibling re-derivations. Downstream `af` trees import it as a validated external instead of re-deriving the triangle step per node.

10 Weighted minimum bound

Lemma 9 (Registry contract for `lem-weighted-min`). *Weighted minimum bound: let $p_1, \dots, p_m$ be positive reals with $\sum_i p_i = 1$ and let $n_1, \dots, n_m$ be real numbers; then $\min$ over i in $\{1, \dots, m\}$ of $n_i \leq \sum_i p_i * n_i$.*

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records an 8-node adversarial tree, root `validated`, taint 8/8 `clean` (the build run's verifier calls were blanked by a network outage and the verify phase was resumed cleanly). The export is `proofs/lem-weighted-min/export.md`.

Role in the chain. Second factored dependency of 10: the index-selection step (some support point has value at most the weighted average). Textbook-adjacent, but registered as its own shard so `af` trees can import it as a validated external instead of re-deriving it across siblings.

11 Zero-sum fan payment

Lemma 10 (Registry contract for lem-fan-payment). *Zero-sum fan payment: let $(w_1, p_1), \dots, (w_m, p_m)$ be a finite family with vectors w_i in $\mathbb{R}^d$ and weights $p_i > 0$ satisfying $\sum_i p_i = 1$, such that every w_i has coordinate sum zero ($\sum_l w_{i,l} = 0$) and the weighted barycenter is zero ($\sum_i p_i w_i = 0$); write $n(w) = \sum_l \max(-w(l), 0)$; then min over i^* in $\{1, \dots, m\}$ of $\sum_i p_i n(w_i - w_{i^*}) \leq 2 * \sum_i p_i n(w_i)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 15-node adversarial tree, root **validated**, taint 15/15 clean, closed in run 3 on the factored workspace; it imports the af-validated dependencies 8 and 9 as externals. Runs 1–2, before the factoring, balloon-aborted at 39 and 47 nodes; the factoring cured the balloon (15 versus 47). The export is **proofs/lem-fan-payment/export.md**.

Role in the chain. The all-mass fan payment lemma is the discrete mechanism behind the certified plateau-2 constant on the reduced fan families of the arm-A campaign: in a reduced edge-fan chart the anchor-pivot score is exactly the weighted average $\sum_i p_i n(w_i - w_*)$, so this lemma bounds the fan-template Φ/δ by 2. The registry shard body additionally records an exact refutation certificate for the over-broad variant (arbitrary w_0 , no zero-sum hypothesis, ratio 5); that certificate is wave material and is not elevated here.

12 Negative-part subadditivity

Lemma 11 (Registry contract for `lem-negpart-subadditive`). *Negative-part subadditivity:*
for all vectors x and y in $\mathbb{R}^d$, writing $n(w) = \sum_l \max(-w(l), 0)$, one has $n(x + y) \leq n(x) + n(y)$.

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 16-node adversarial tree, root `validated`, taint 16/16 clean, closed in run 1. The export is `proofs/lem-negpart-subadditive/export.md`. The lemma was pre-factored *before* elevation, applying the balloon lesson from `lem-fan-payment` instead of re-learning it.

Role in the chain. Pre-factored dependency of 12: the D-restricted fan proof's barycenter step $D \geq n(\sum_{i \in A} p_i w_i)$ uses exactly this pointwise subadditivity (with nonnegative homogeneity, which is definitional).

13 D-restricted zero-sum fan payment

Lemma 12 (Registry contract for lem-fan-payment-restricted). *D-restricted zero-sum fan payment: let $(w_1, p_1), \dots, (w_m, p_m)$ be a finite family with vectors w_i in $\mathbb{R}^d$ and weights $p_i > 0$ satisfying $\sum_i p_i = 1$, such that every w_i has coordinate sum zero and the weighted barycenter is zero ($\sum_i p_i w_i = 0$); write $n(w) = \sum_l \max(-w(l), 0)$; let w_* be a minimizer of $v \rightarrow \sum_i p_i n(w_i - v)$ over $\{w_1, \dots, w_m\}$ and let $A = \{i : n(w_i - w_*) > 0\}$; then $\sum_{i \in A} p_i n(w_i - w_*) \leq (2 + \sqrt{2}) * \sum_{i \in A} p_i n(w_i)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 27-node adversarial tree, root **validated**, taint 27/27 clean, closed in run 1 after ten rounds; it imports the af-validated dependencies 8 and 11 as externals. The export is **proofs/lem-fan-payment-restricted/export.md**. Proof shape: with q the duplicate-pivot mass, horn (i) gives $N \leq D/q$ and horn (ii) gives $N \leq 2(1 - q)/(1 - 2q) D$ for $q < 1/2$; the bounds cross at $q_0 = 1 - 1/\sqrt{2}$ with common value $2 + \sqrt{2}$.

Role in the chain. The D-restricted refinement of 10: the denominator keeps only rows with positive distance to the selected pivot, the fan shadow of the degenerate-set restriction behind **conj-degenerate-payment**. The registry shard body records exact certificates (wave material, not elevated here) refuting constant 2 for this variant and attaining $2 + \sqrt{2}$ in the limit, so the validated constant is sharp. The remaining obstruction toward **conj-degenerate-payment** is the lift from $\theta^{-\frac{1}{2}}$ argmin rows to this fan model.

14 Pivot-removing max-stationarity

Lemma 13 (Registry contract for lem-pivot-removing-move).

Pivot-removing max-stationarity:

let P be a rank-3 exact signed idempotent (square real matrix with $P^2 = P$ and all row sums equal to 1), let $U = (u_0, u_1, u_2)$ be an actual-row chart whose rows $p_{\{u_0\}}$, $p_{\{u_1\}}$, $p_{\{u_2\}}$ form a basis of the row space, let $\text{Vol_max}(P)$ be the maximum Gram volume over actual-row charts, $m_U = \text{Vol}(U)/\text{Vol_max}(P)$, define old coordinates $a_t(i)$ by $p_i = \sum_t a_t(i)p_{\{u_t\}}$, $\beta_r(i) = P_{\{u_r\}i}$, $\lambda_r(i) = 1 - a_r(i)$, $\mu_r(i) = \sum_{t \neq r} \max(-a_t(i), 0)$, $E_r(i) = \max(\mu_r(i) - \lambda_r(i), 0)$, $\Phi_r(U) = \sum_i \max(\beta_r(i), 0)E_r(i)$, and $\Phi(U) = \max_r \Phi_r(U)$, assume U is a theta-half Φ -argmin meaning $m_U \geq 1/2$ and $\Phi(U)$ is minimal among actual-row charts W with $\text{Vol}(W)/\text{Vol_max}(P) \geq 1/2$, fix a maximal pivot s with $\Phi_s(U) = \Phi(U)$, and let $j \notin \{u_0, u_1, u_2\}$ have $c = a_s(j) \neq 0$ such that the pivot-removing chart $V_j = U - u_s + j$ is theta-half admissible, equivalently $|a_s(j)| * m_U \geq 1/2$ with volume factor $\text{Vol}(V_j)/\text{Vol}(U) = |a_s(j)|$; defining new coordinates on V_j by $a_s^j(i) = a_s(i)/c$ and $a_t^j(i) = a_t(i) - a_s(i)a_t(j)/c$ for $t \neq s$, $E_r^j(i) = \max(\sum_{q \neq r} \max(-a_q^j(i), 0) - (1 - a_r^j(i)), 0)$, $\Psi_j = \Phi_s(V_j) = \sum_i \max(P_{\{ji\}}, 0)E_s^j(i)$, and $\Gamma_j = \max_{r \neq s} \Phi_r(V_j) = \max_{r \neq s} \sum_i \max(P_{\{ur\}i}, 0)E_r^j(i)$, one has $\Phi_s(U) \leq \max(\Psi_j, \Gamma_j)$.

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 9-node adversarial tree, root validated, taint 9/9 clean, closed in run 1 after three rounds with zero challenges. The export is `proofs/lem-pivot-removing-move/export.r`. The statement was codified from the arm-G wave record with a fully-inline contract and elevated pre-factored, so the tree stayed small.

Role in the chain. This is the minimality tool of the argmin engine: it is the first statement of the campaign that genuinely uses Φ -minimality of the chart, via the pivot-removing move with exact volume factor $|a_s(j)|$. Every attack on the remaining self-support control (conj-sc and its pivot-removing blocker reduction) consumes this disjunction: an admissible pivot-removing move must be blocked by the new-pivot score Ψ_j or by collateral Γ_j on another pivot. The wave-9 record realizes the volume-inadmissible and Ψ -blocked alternatives by exact certified instances; the collateral alternative is the open branch, and its analysis now rests on this validated statement.

15 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. The first thirteen **af**-validated results are reproduced in 1–13; they are not duplicated in this ledger. The fourteenth and fifteenth (**lem-collateral-import**, **lem-cross-pivot-cancellation**) are anchored below pending their own sections.

Table 1: Registry status ledger for report anchors outside the thirteen validated sections.

Registry id	Status	Contract gloss
conj-degenerate-payment	conjecture	Pivot-local weighted near-degenerate payment bound; the payment horn of the (EX) argmin charge, exact-tested on the wave zoo but unproved.
conj-degenerate-transport	conjecture	Degenerate transverse transport bound; the remaining payment-horn gap after the wave-12 lift, exact-tested but unproved.
conj-ex	conjecture	(EX) chart-bound working form for exact signed idempotents; still a conjectural input equivalent to conj-kernel in the registry.
conj-kernel	conjecture	Kernel Conjecture, the geometric input intended to close op-exposed-hull and hence op-classical .
conj-no-free-frontier	conjecture	Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.
conj-rh	conjecture	Repaired orphan horn: post-fan orphan demand bounded by the unified class/signed/own-negativity budget plus fan residual; exact floor four on the coefficient.
conj-sc	conjecture	Argmin self-support/cancellation control for non-fan beta-positive rows; reduced to the pivot-removing blocker theorem, collateral branch open.
conj-skinny-shadow-cap	conjecture	Skinny two-shadow cap at the root-delta scale, the corrected Route-B statement superseding the retired dual-localization contract.
ex-hume	proved-mod-audit	Explicit 3-by-3 sharpness family for the exponent in op-classical ; inherited proof is not rigorous here.
lem-canonical-separator	proved-mod-audit	Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.
lem-dual-localization	obstruction (superseded)	Retired 2026-07-04: the transcribed contract is a distance tautology; superseded by conj-skinny-shadow-cap .
lem-exposed-circuit	proved-mod-audit	Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.
lem-collateral-import	proved (af: validated)	Collateral-import bound (CI) for the pivot-removing move, sharp on the wave-10 witness; af-validated in-repo 2026-07-04 (32-node tree, taint clean).
lem-cross-pivot-cancellation	proved (af: validated)	Cross-pivot B-L cancellation identity behind the import anatomy; af-validated in-repo 2026-07-04 (23-node tree, taint clean); near-definitional.
lem-import-reduction	proved-mod-audit	Universal reduction of the collateral import to the cross-pivot masses; wave paper-proof pending elevation.

lem-leakage	proved-mod-audit	Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.
lem-wiggle-rigidity	proved-mod-audit	F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.
obs-deep-leakage	heuristic	Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.
obs-fwr-gap	heuristic	F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.
obs-linear-law-finite-delta	numerical	Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.
obs-orphan-amplifier	proved-mod-audit	Exact two-orphan family killing every class/signed-only orphan budget; repaired ratio tends to four; wave paper-proof, orchestrator-recomputed.
obs-sigma-halo-nonrobust	numerical	Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.
op-classical	open	North-star classical projection stability problem; still open in this repo.
op-exposed-hull	open	Global exposed-hull lemma that would imply op-classical through the inherited factorization route; open.
prop-approx-simplex	proved-mod-audit	Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.
thm-classical-factorization	proved-mod-audit	Conditional classical factorization theorem through cluster geometry and op-exposed-hull ; not rigorous here.
thm-cluster	proved-mod-audit	Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.
thm-corner-constants	proved-mod-audit	Corner-scale constants, with numerical components explicitly not promoted.
thm-rank-one	proved-mod-audit	Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only.
thm-simplex	proved-mod-audit	Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.
thm-well-exposed	proved-mod-audit	Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.
